8 Programming

8.1 Programming concepts

Candidates should be able to:	Notes and guidance
1 Declare and use variables and constants	
2 Understand and use basic data types	• Including: – integer – real – char – string – Boolean
3 Understand and use input and output	
4 (a) Understand and use the concept of sequence	
(b) Understand and use the concept of selection	• Including: – `IF` statements – `CASE` statements
(c) Understand and use the concept of iteration	• Including: – count-controlled loops – pre-condition loops – post-condition loops
(d) Understand and use the concepts of totalling and counting	
(e) Understand and use the concept of string handling	• Including: – length – substring – upper – lower • The first character of the string can be position zero or one

8.1 Programming concepts continued

Candidates should be able to:	Notes and guidance
(f) Understand and use arithmetic, relational and logical operators	• Arithmetic, limited to: – `+` – `-` – `/` – `*` – `^` (raised to power of) – `MOD` – `DIV` • Relational, limited to: – `=` – `<` – `<=` – `>` – `>=` – `<>` (not equal to) • Logical, limited to: – `AND` – `OR` – `NOT`
5 Understand and use nested statements	• Including nested selection and iteration • Candidates will **not** be required to write more than three levels of nested statements
6 (a) Understand what is meant by procedures, functions and parameters (b) Define and use procedures and functions, with or without parameters (c) Understand and use local and global variables	• Procedures and functions may have up to three parameters
7 Understand and use library routines	• Including: – MOD – DIV – ROUND – RANDOM

8.1 Programming concepts continued

Candidates should be able to:	Notes and guidance
8 Understand how to create a maintainable program	• Including appropriate use of: – meaningful identifiers – the commenting feature provided by the programming language – procedures and functions – relevant and appropriate commenting of syntax • Use meaningful identifiers for: – variables – constants – arrays – procedures and functions

8.2 Arrays

Candidates should be able to:	Notes and guidance
1 Declare and use one-dimensional (1D) and two-dimensional (2D) arrays	
2 Understand the use of arrays	• Including the use of variables as indexes in arrays
3 Write values into, and read values from, an array using iteration	• The first index can be zero or one • Including nested iteration

8.3 File handling

Candidates should be able to:	Notes and guidance
1 Understand the purpose of storing data in a file to be used by a program	
2 Open, close and use a file for reading and writing	• Including: – read and write single items of data – read and write a line of text